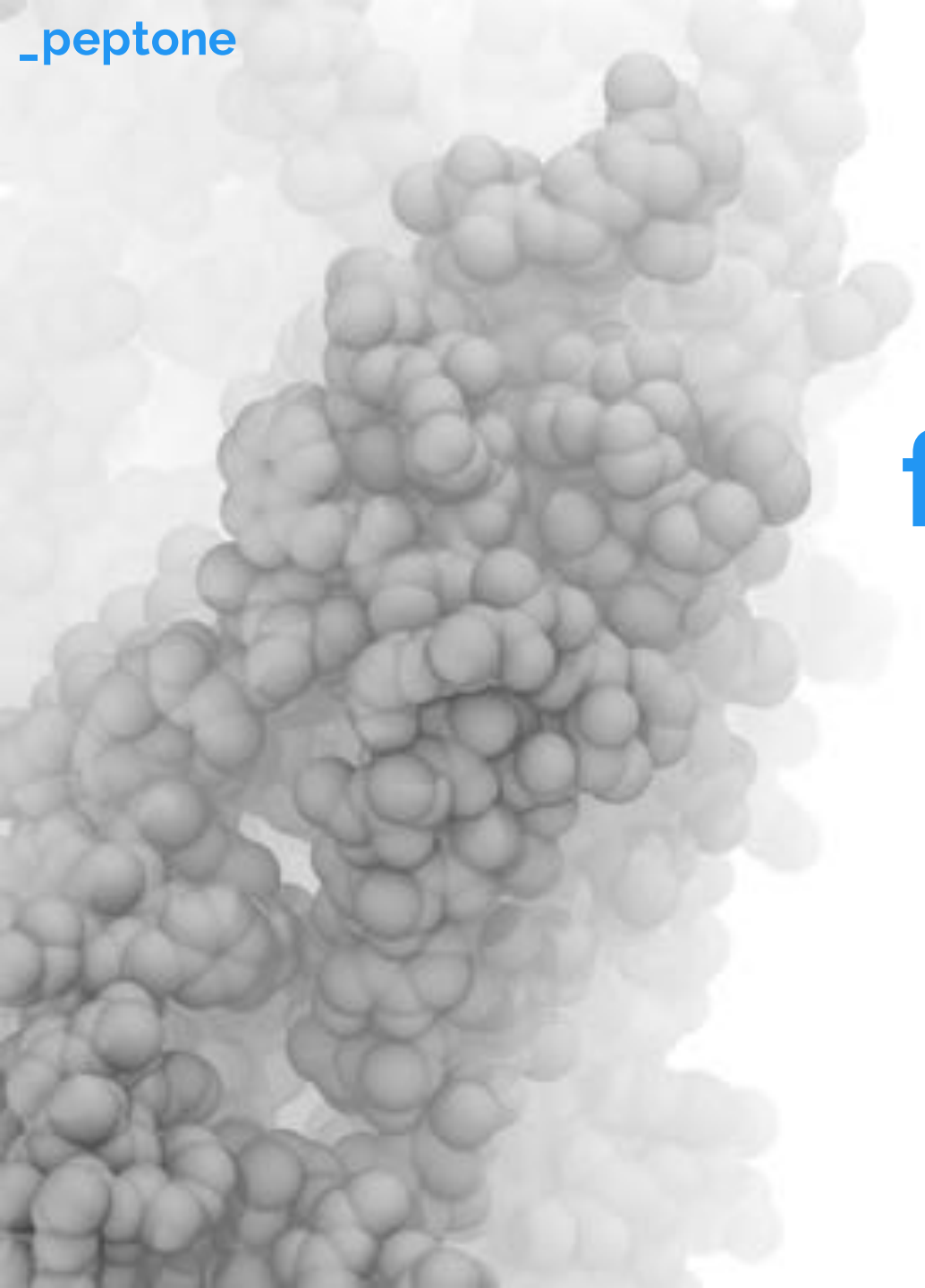
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Protein biophysics from Machine Learning and AI perspective

Kamil Tamiola

_peptone – The Protein Intelligence Company

Talk 23058

NVIDIA GTC Europe

Proteins are building blocks of life

20% of the dry weight of human cells

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Food



Spirits



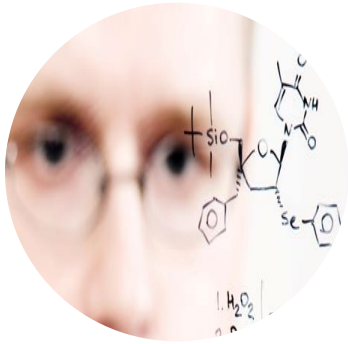
Materials



Biofules



Pharmaceuticals

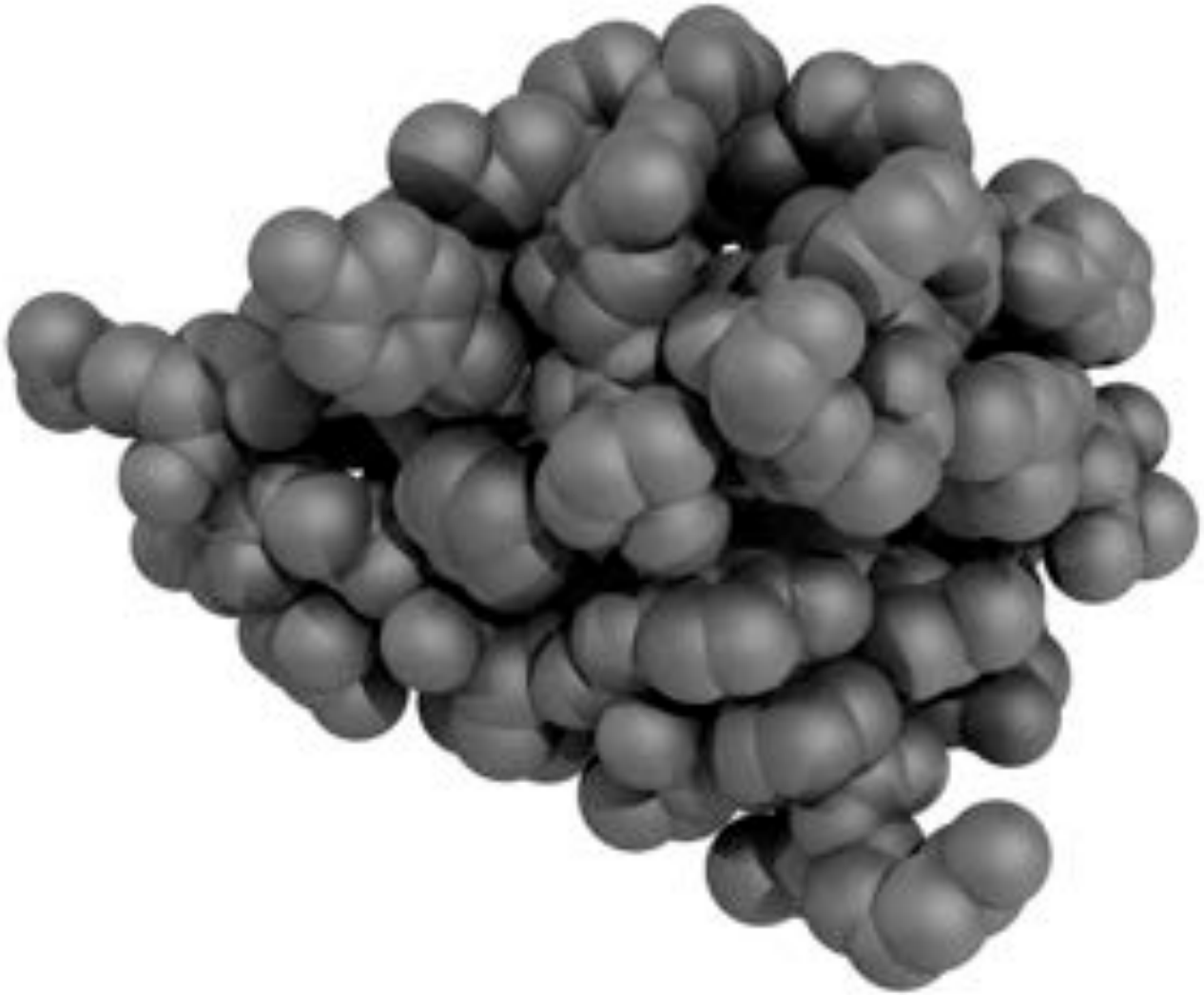


Biosimilars

800 billion USD

https://www2.deloitte.com/us/en/pages/outlooks/industry-outlooks.html

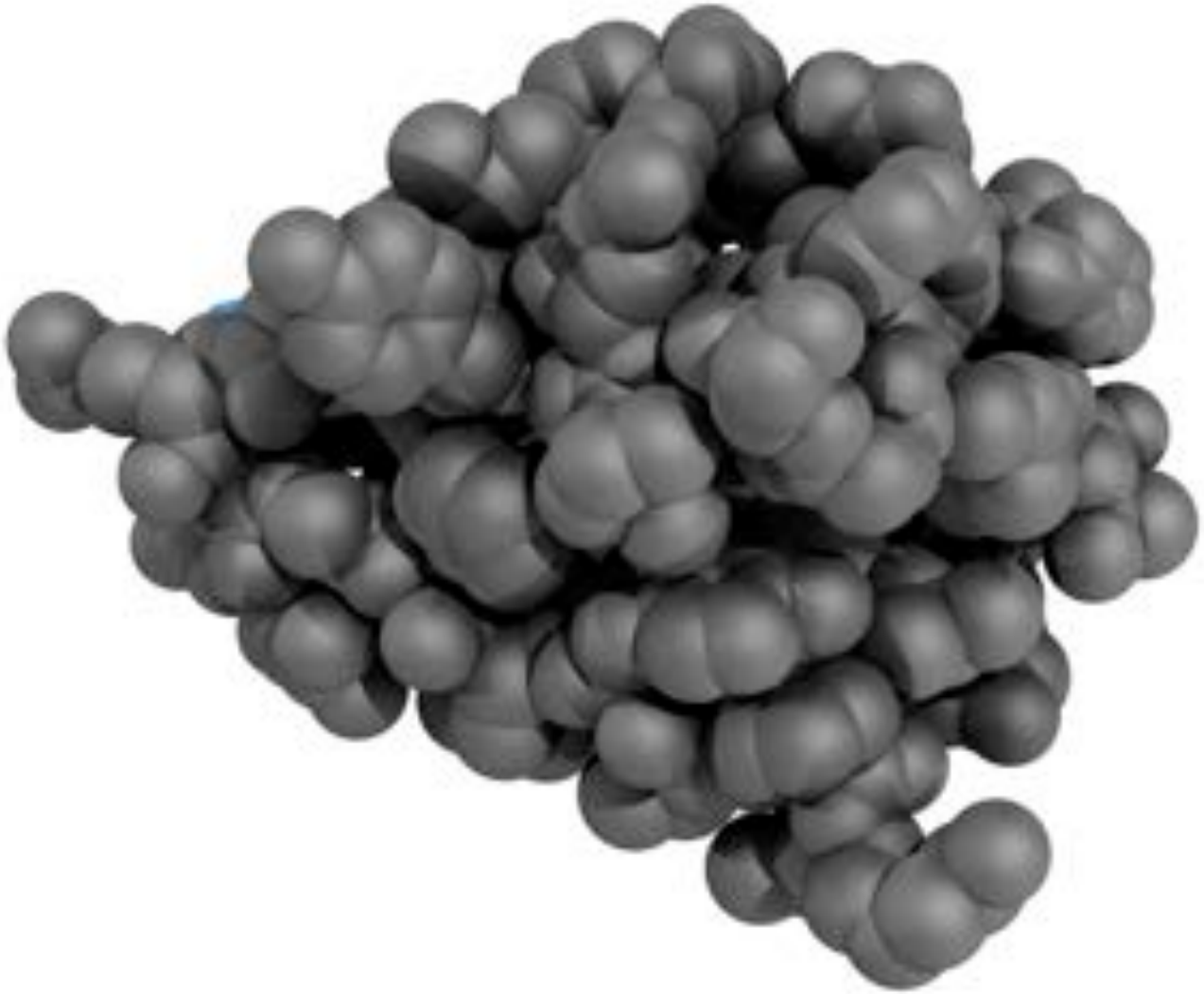
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Human Insulin

The most successful protein therapeutic ever

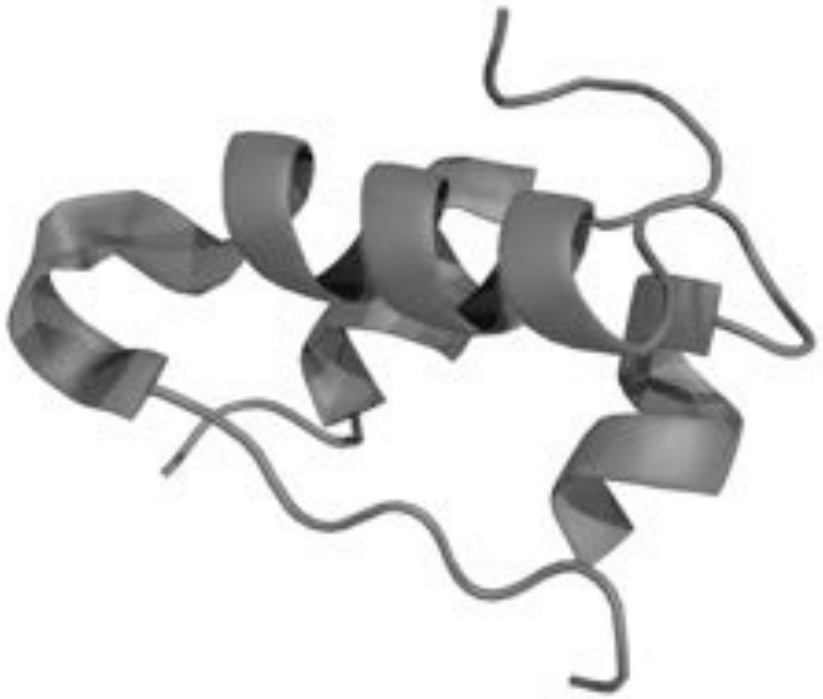
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Human Insulin

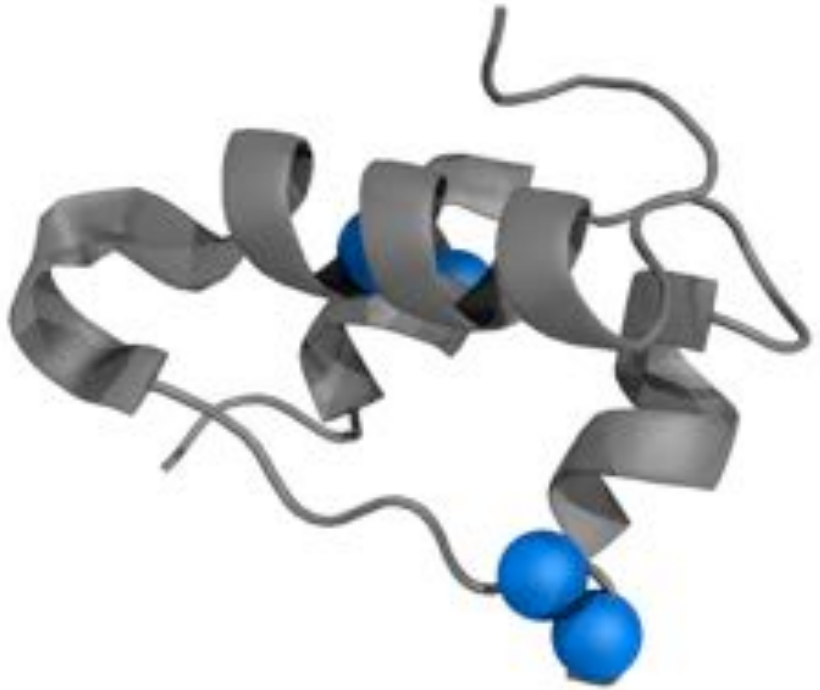
discovered in 1921
characterized in 1958
large scale production in 1982

_peptone



Human

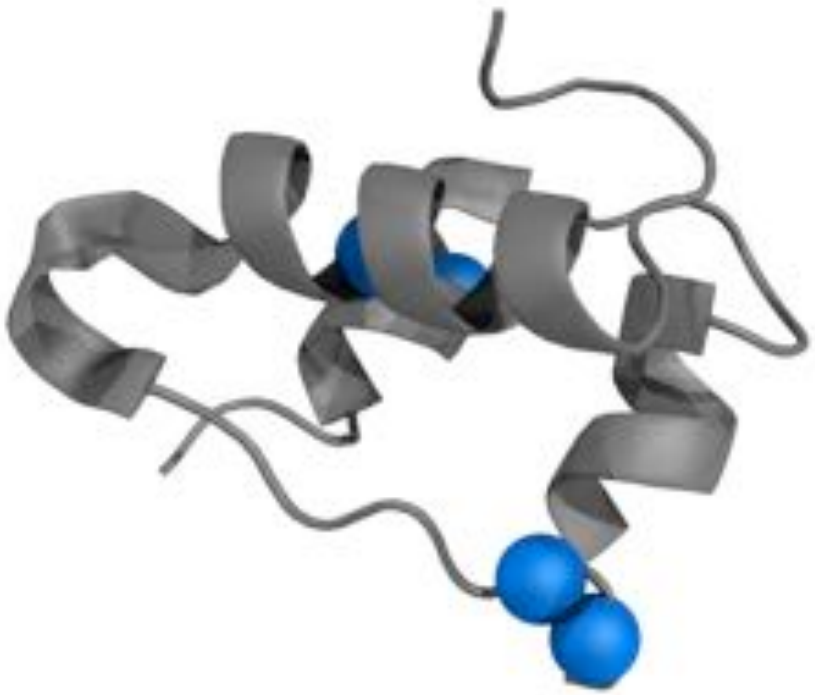
FVNQHLCGSHLVEALYLVCGERGFFYTPKTA
VEQCCTSICSLYQLENYCN



Analog

FVNQHLCGSHLVEALYLVCGERGFFYT**KP**TA
VEQCCTSICSLYQL**E**NYCN

_peptone



Humalog®

FVNQHLCGSHLVEALYLVCGERGFFYT**KP**TA
VEQCCTSICSLYQL**E**NYCN



_peptone

Wild type

FVNQHLCGSHLVEAL
YLVCGERGFFYTPKTA
VEQCCTSICSLYQLEN
YCN

30 years



150B $

Analog

FVNQHLCGSHLVEALY
LVCGERGFFYT**KP**TA
VEQCCTSICSLYQL**E**N
YCN

Why did it take so long?

_peptone

Wild type

FVNQHLCGSHLVEALY
LVCGERGFFYTPKTA
VEQCCTSICSLYQLENY
CN

Combinatorial mutagenesis →

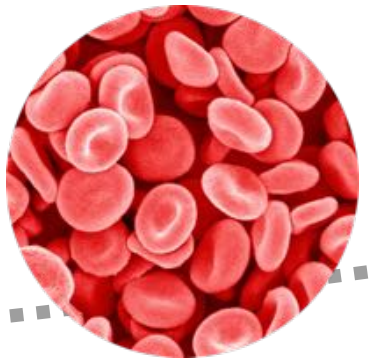
20^{58}

Mutant combinations

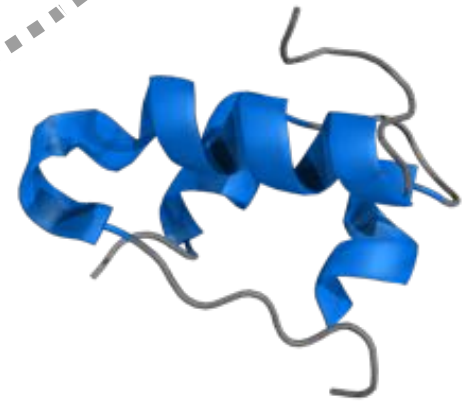
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Scale and complexity









Complexity

10^{10}
people
Population of earth

10^{13}
cells
Cells in human body

10^{24}
seconds
Age of Universe

$\sim 10^{58}$
combinations
Number of mutants

Scale and complexity

Protein engineering is hampered by combinatorial complexity

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Journal List › J Clin Invest › v.118(6); 2008 Jun 2 › PMC2350430

JCI The Journal of Clinical Investigation
Published by The American Society for Clinical Investigation | Founded 1908

J Clin Invest. 2008 Jun 2; 118(6): 2148–2156.
Published online 2008 May 1. doi: 10.1172/JCI33777
PMCID: PMC2350430

Seven mutations in the human insulin gene linked to permanent neonatal/infancy-onset diabetes mellitus

Carlo Colombo,[1] Ottavia Porzio,[2] Ming Liu,[3] Ornella Massa,[1] Mario Vasta,[4] Silvana Salardi,[5] Luciano Beccaria,[6] Carla Monciotti,[7] Sonia Toni,[8] Oluf Pedersen,[9] Torben Hansen,[9] Luca Federici,[10] Roberta Pesaventio,[11] Francesco Cadario,[12] Giorgio Federici,[2] Paolo Ghirri,[13] Peter Arvan,[3] Dario Iafusco,[14] and Fabrizio Barbetti[1,2,15]

the Early Onset Diabetes Study Group of the Italian Society of Pediatric Endocrinology and Diabetes (SIEDP)

[1]Laboratory of Molecular Endocrinology and Metabolism, Bambino Gesù Children's Hospital, Scientific Institute (IRCCS), Rome, Italy. [2]Department of Internal Medicine, University of Tor Vergata, Rome, Italy. [3]Division of Metabolism, Endocrinology, and Diabetes, Department of Internal Medicine, University of Michigan Health System, Ann Arbor, Michigan, USA. [4]Diabetes Clinic, Urbino City Hospital, Urbino, Italy. [5]Department of Pediatrics, University of Bologna, Bologna, Italy. [6]Department of Pediatrics, Alessandro Manzoni Hospital, Lecco, Italy. [7]Department of Pediatrics, University of Padua, Padua, Italy. [8]Regional Center for Juvenile Diabetes, Meyer Pediatric Hospital, Florence, Italy. [9]Steno Diabetes Center, Gentofte, Denmark. [10]Ce.S.I. Center for Studies on Aging, G. d'Annunzio University Foundation and Department of Biomedical Sciences, University of Chieti G. d'Annunzio, Chieti, Italy. [11]Pediatric Unit, Boldrini Hospital, Thiene, Italy. [12]Department of Pediatrics, University of Eastern Piedmont, Interdisciplinary Research Center of Autoimmune Diseases, Novara, Italy. [13]Department of Pediatrics, Division of Neonatology, University of Pisa, Pisa, Italy. [14]Department of Pediatrics, 2nd University of Naples, Naples, Italy. [15]Biomedical Park Foundation S. Raffaele, Rome, Italy.

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J Clin Invest

Journal List › HHS Author Manuscripts › PMC4833475

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Circ Res. Author manuscript; available in PMC 2017 Apr 1.
Published in final edited form as:
Circ Res. 2016 Apr 1; 118(7): 1151–1169.
doi: 10.1161/CIRCRESAHA.116.306206
PMCID: PMC4833475
NIHMSID: NIHMS765147

Insulin Signaling and Heart Failure

Christian Riehle, MD, PhD[1] and E. Dale Abel, MB.BS., D.Phil.

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Journal List › J Diabetes Investig › v.2(2); 2011 Apr 7 › PMC4015537

JDI Journal of Diabetes Investigation
Official Journal of the Asian Association for the Study of Diabetes

J Diabetes Investig. 2011 Apr 7; 2(2): 92–100.
Published online 2011 Feb 17. doi: 10.1111/j.2040-1124.2011.00100.x
PMCID: PMC4015537

Insulin gene mutations and diabetes

Masahiro Nishi[1] and Kishio Nanjo[*2]

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Journal List › HHS Author Manuscripts › PMC4253975

HHS Public Access
Author manuscript
Peer-reviewed and accepted for publication
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Neurobiol Dis. Author manuscript; available in PMC 2015 Dec 1.
Published in final edited form as:
Neurobiol Dis. 2014 Dec; 72PA: 92–103.
Published online 2014 Sep 16. doi: 10.1016/j.nbd.2014.09.001
PMCID: PMC4253975
NIHMSID: NIHMS633774

Insulin Resistance in Alzheimer's Disease

Kelly T Dineley,[1] Jordan B Jahrling,[1,*] and Larry Denner[2,#]

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The publisher's final edited version of this article is available at Neurobiol Dis
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Mutations can be dangerous

Combinatorial mutagenesis must exclude cases that lead to pathologies

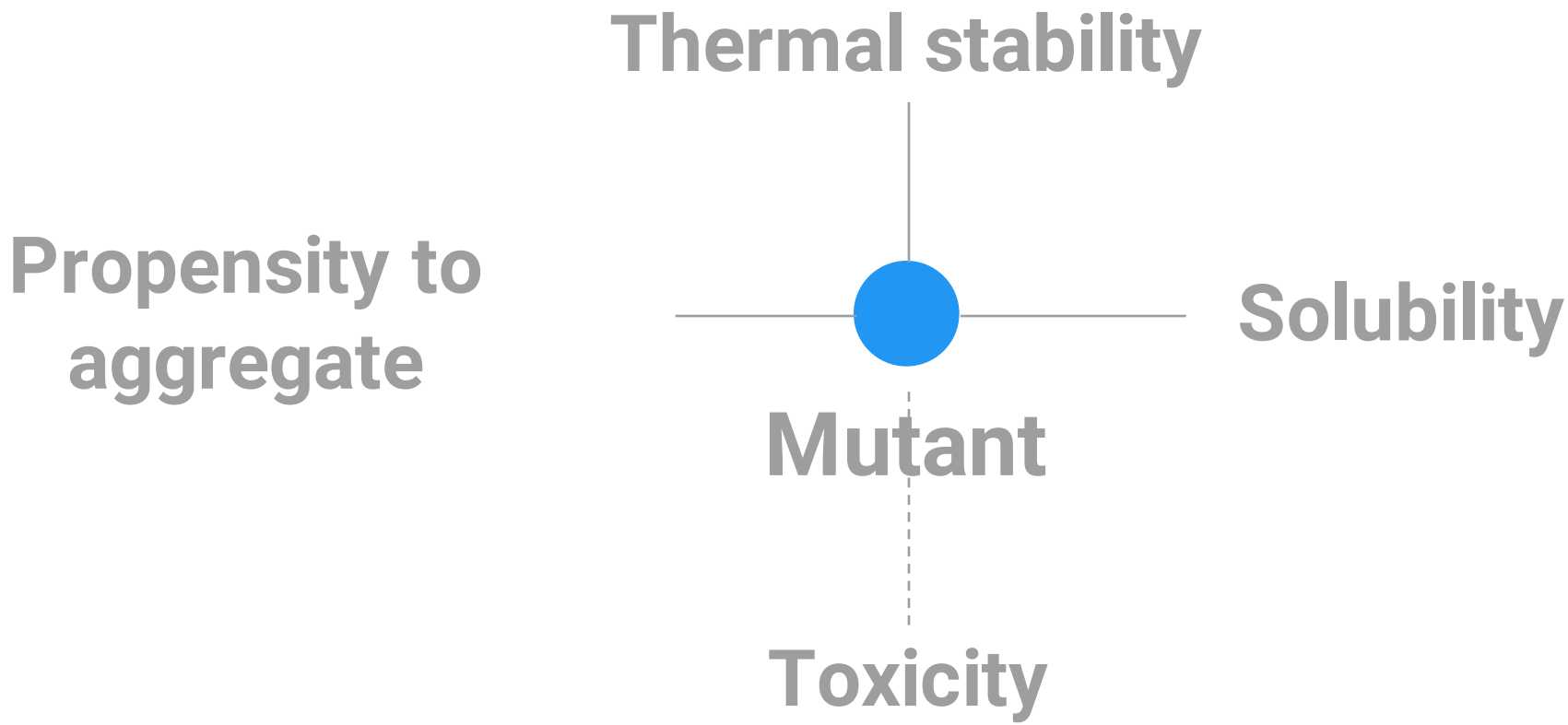
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How can we use AI to accelerate insulin design?

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Assign properties

Every mutant is a point with properties



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Combinatorial space

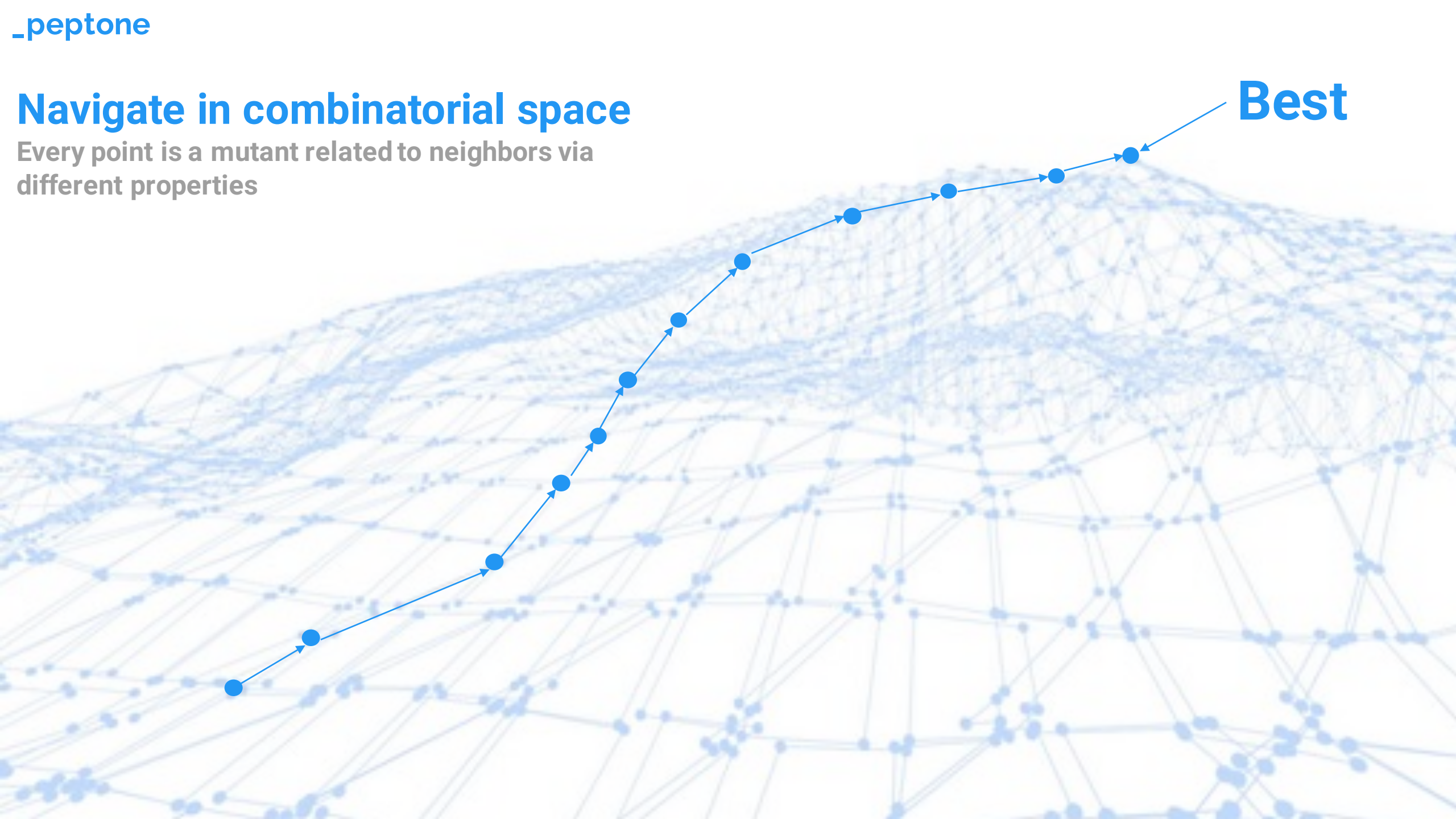
Every point is a mutant related to neighbors via different properties



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Navigate in combinatorial space

Every point is a mutant related to neighbors via different properties



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Physics

Bioinformatics
doi.10.1093/bioinformatics/xxxxxx
Advance Access Publication Date: Day Month Year
Manuscript Category

OXFORD

High Performance Computing

pepKalc - scalable and comprehensive calculation of electrostatic interactions in random coil polypeptides

Kamil Tamiola [1,2*], Ruud M. Scheek [2], Pieter van der Meulen [2], and Frans A.A. Mulder [2,3,*]

[1] Peptone - The Protein Intelligence Company, Amsterdam 1001AM, The Netherlands
[2] Groningen Biotechnology Institute, University of Groningen, Groningen 9747AG, The Netherlands
[3] Department of Chemistry and Interdisciplinary Nanoscience Center INANO, Aarhus 8000, Denmark

*To whom correspondence should be addressed.

Associate Editor: XXXXXXX

Received on XXXXX; revised on XXXXX; accepted on XXXXX

Abstract

Motivation: Polypeptide sequence length is the single dominant factor hampering the effectiveness of currently available software tools for *de novo* calculation of amino acid-specific protonation constants in disordered polypeptides.

Results: We have developed **pepKalc** , a robust simulation software for the comprehensive evaluation of protein electrostatics in unfolded state. Our software completely removes the limitations of the previously reported Monte-Carlo approaches in the computation of protein electrostatics, by using a hybrid approach that effectively combines exact and mean-field calculations to rapidly obtain accurate results. Paired with a modern architecture GPU, **pepKalc** is capable of evaluating protonation behavior for an arbitrary-size polypeptide in a sub-second time regime.

Knowledge

Structural propensity database of proteins for the next generation machine learning problems

Kamil Tamiola[1,*], Matthew M. Heberling[1], and Jan Domański[2]

[1] Peptone - The Protein Intelligence Company, Amsterdam, Hullenbergweg 280, 1101BV, The Netherlands
[2] Department of Biochemistry, University of Oxford, South Parks Road, Oxford OX1 3QU, United Kingdom
[*] Correspondence should be addressed to: kamil@peptone.io

ABSTRACT

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Introduction

Protein sequence likely holds the key to a perplexing mystery in the life sciences–the protein folding problem[1]. Therefore, immense efforts have been devoted to unraveling the sequence-structure relationship in polypeptides[2–4]. Although the fundamental forces of protein folding are known[5–7], complexity has hampered development of accurate folding prediction tools[8]. ML offers hope for protein structure prediction[9,10]. However, the quality of input data will dictate the broad applicability of ML-based prediction tools, which are currently hindered through the disregard of two fundamentals factors: (1) intrinsic protein disorder at the residue level and (2) experimental conditions. For the latter, fundamental laws of equilibrium

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Bioinformatics
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Advance Access Publication Date: Day Month Year
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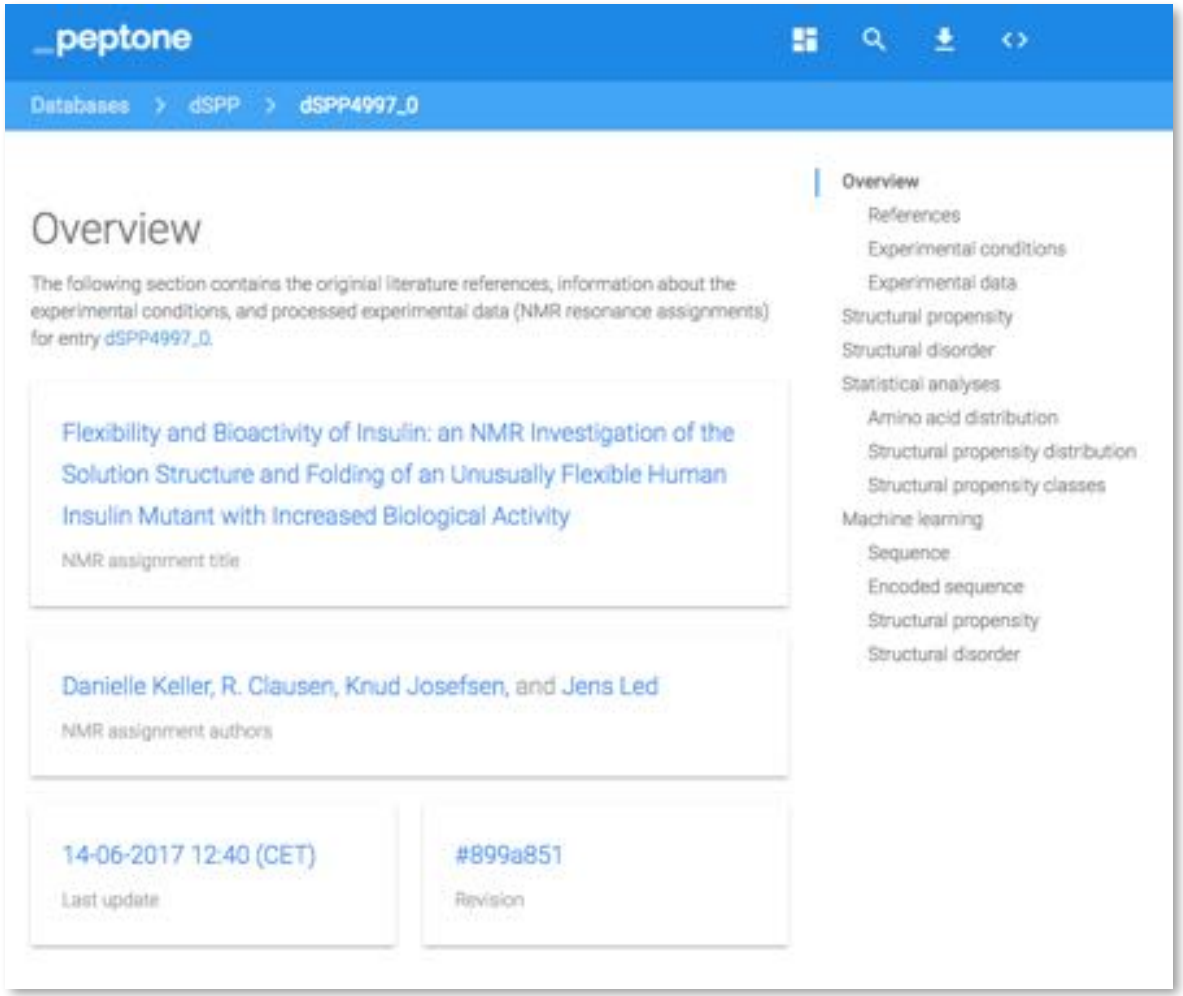
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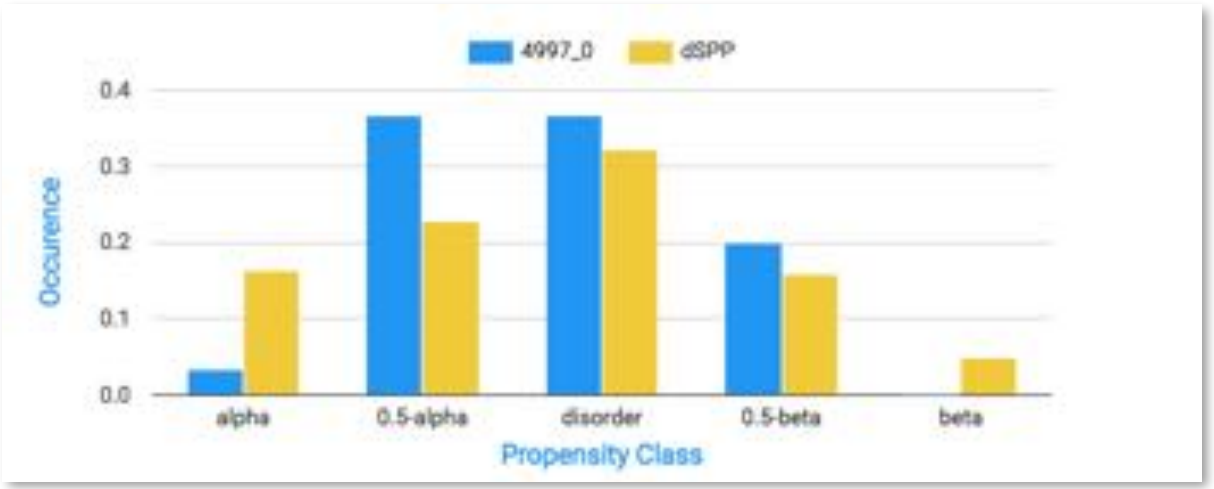
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Database of protein order and disorder

7200+ medically relevant proteins in Machine Learning format





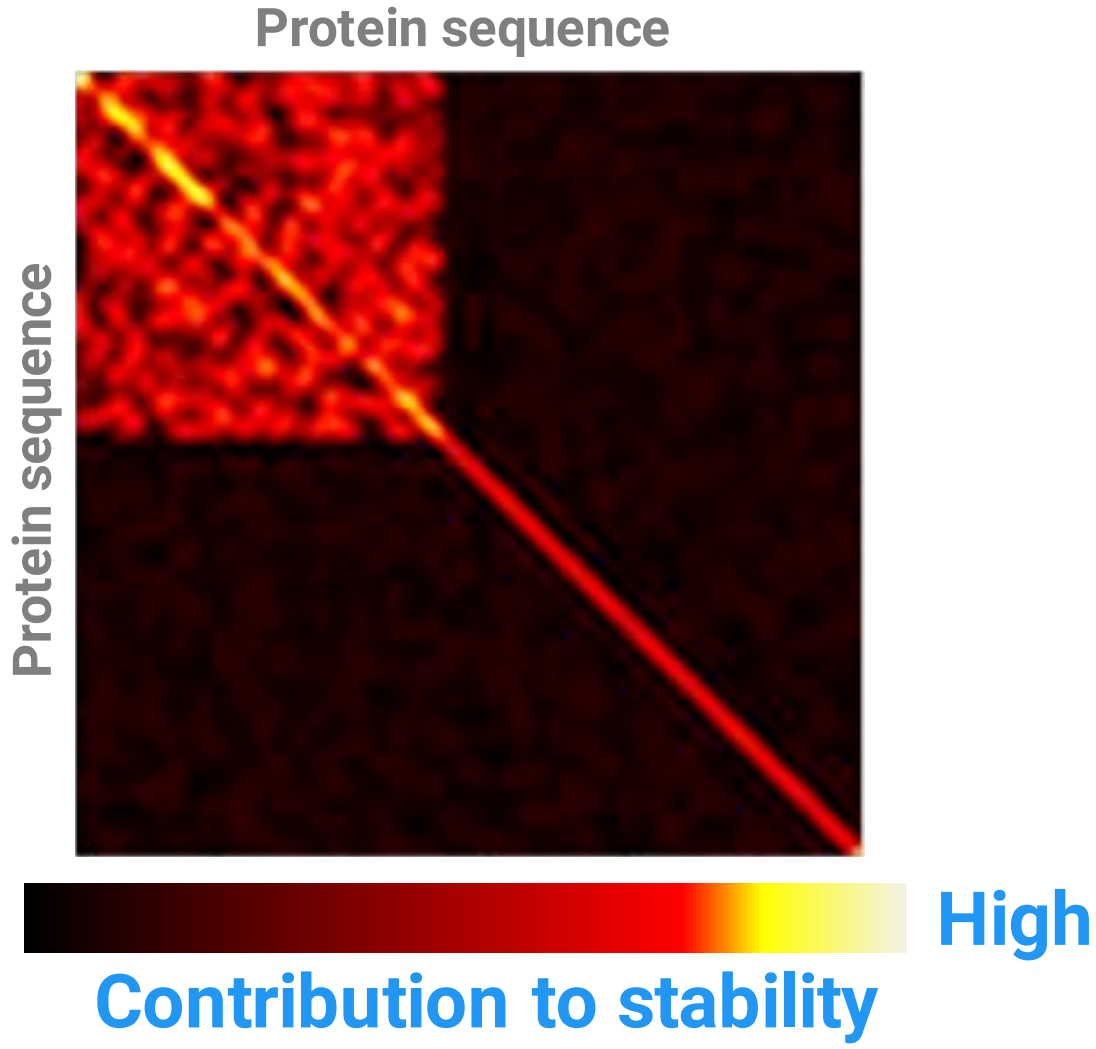


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Predict the effect of mutations on stability

Predictor trained using freeware dspp-keras* library





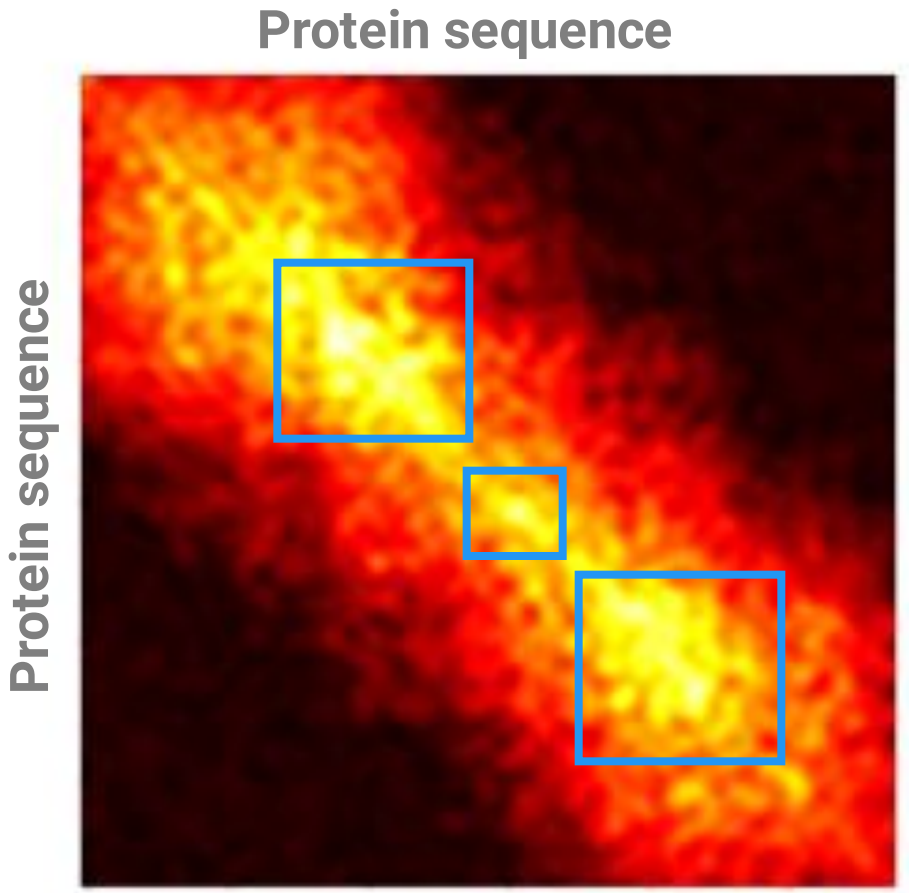
*https://peptone.io/dspp

_peptone

Select residues

FVNQHLCG**SHLVE**ALY**LVCG**
ERGFF**YTPK**TA

GIVEQCCTSICS**LYQ**L**E**NYCN



Residual disorder

Protein mutants can be automatically
selected according to the degree of **disorder**

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20^{58} → 20^{17}

Total number of mutants to screen

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Physics

Bioinformatics
doi.10.1093/bioinformatics/xxxxxx
Advance Access Publication Date: Day Month Year
Manuscript Category



High Performance Computing

pepKalc - scalable and comprehensive calculation of electrostatic interactions in random coil polypeptides

Kamil Tamiola [1,2*], Ruud M. Scheek [2], Pieter van der Meulen [2], and Frans A.A. Mulder [2,3,*]

[1] Peptone - The Protein Intelligence Company, Amsterdam 1001AM, The Netherlands
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[3] Department of Chemistry and Interdisciplinary Nanoscience Center (iNANO), Aarhus 8000, Denmark

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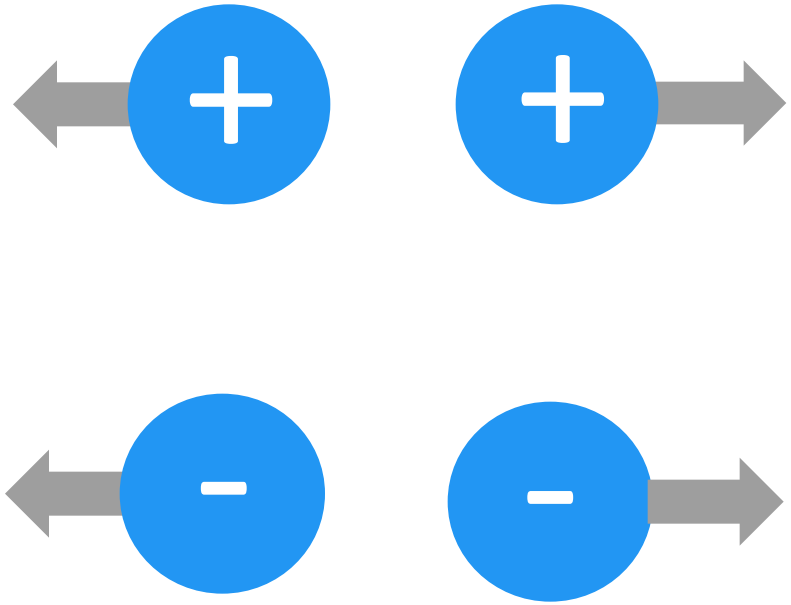
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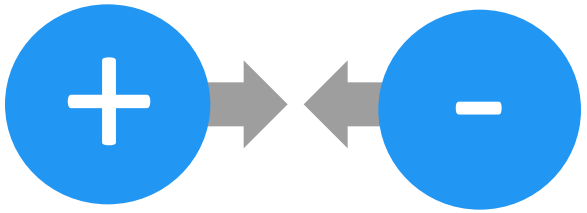
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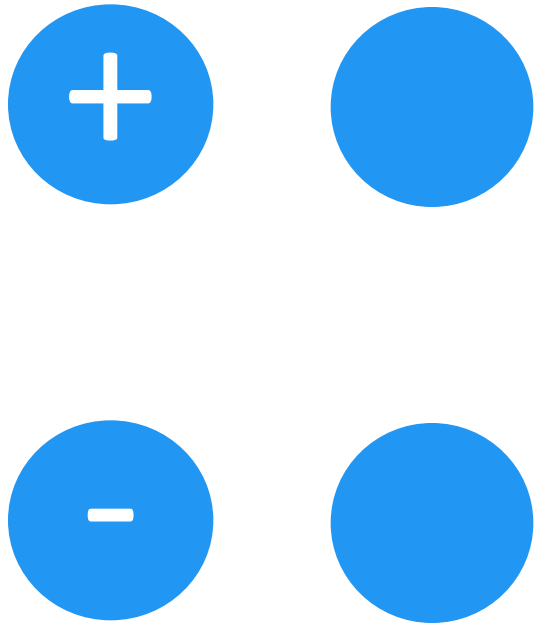
Repulsion



Attraction



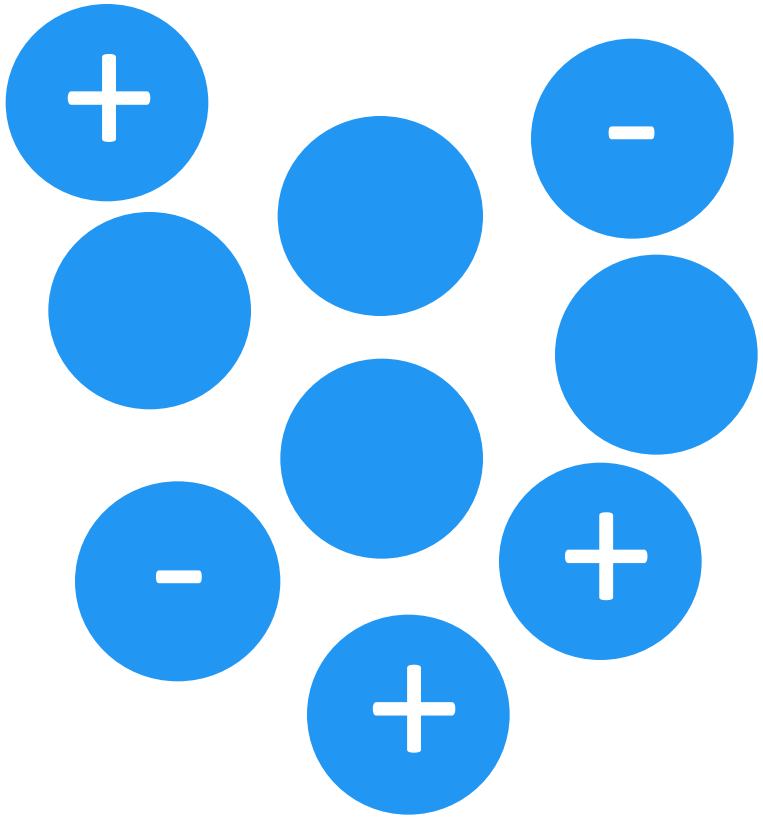
No effect

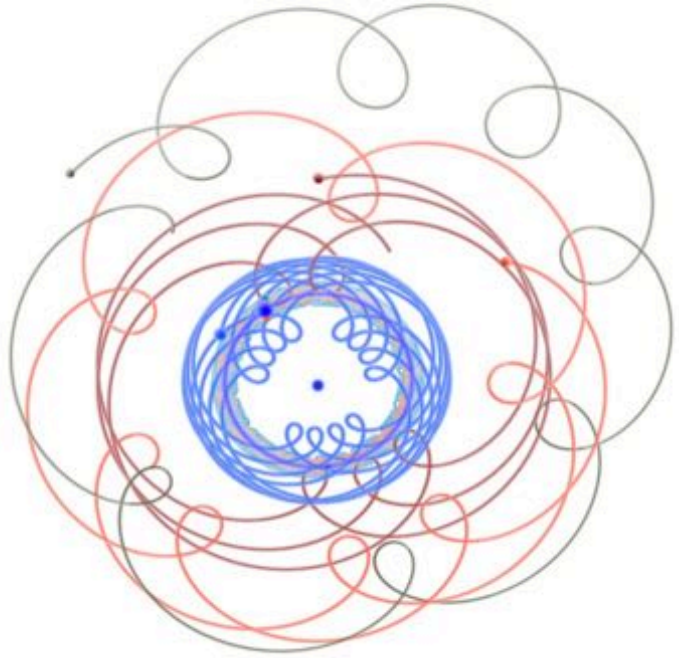


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N-interacting bodies

Conceptually similar to calculating interactions of celestial bodies





Jet Propulsion Laboratory
California Institute of Technology

Charged residues

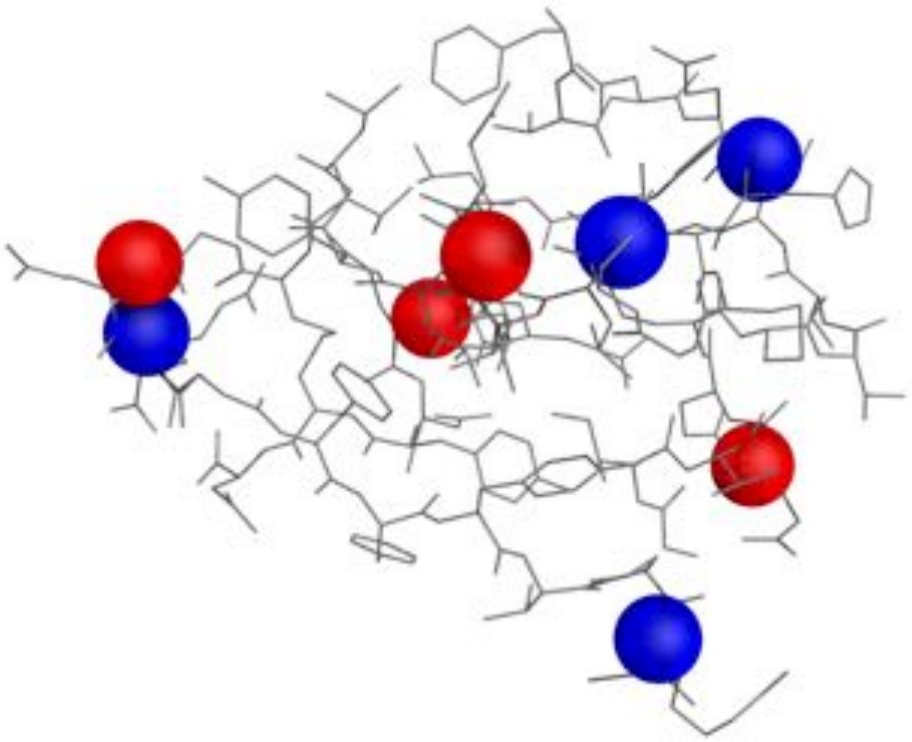
FVNQ**H**LCGS**H**LV**E**AL

YLVCG**E**R**R**GFFYTP**K**TA

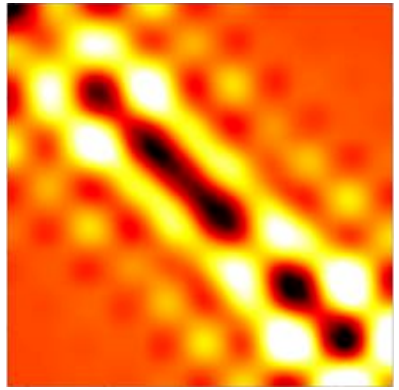
V**E**QCCTSICSLYQL**E**N

YCN

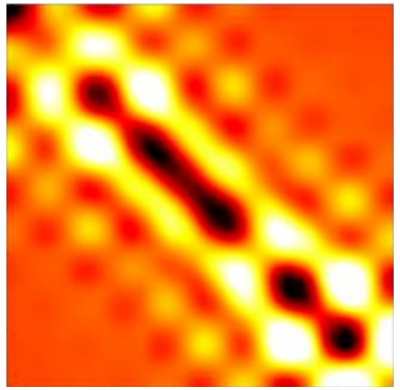
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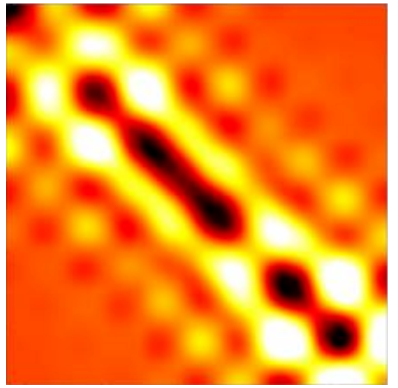
Neutral



Positive



Negative



Low



High

Interaction energy

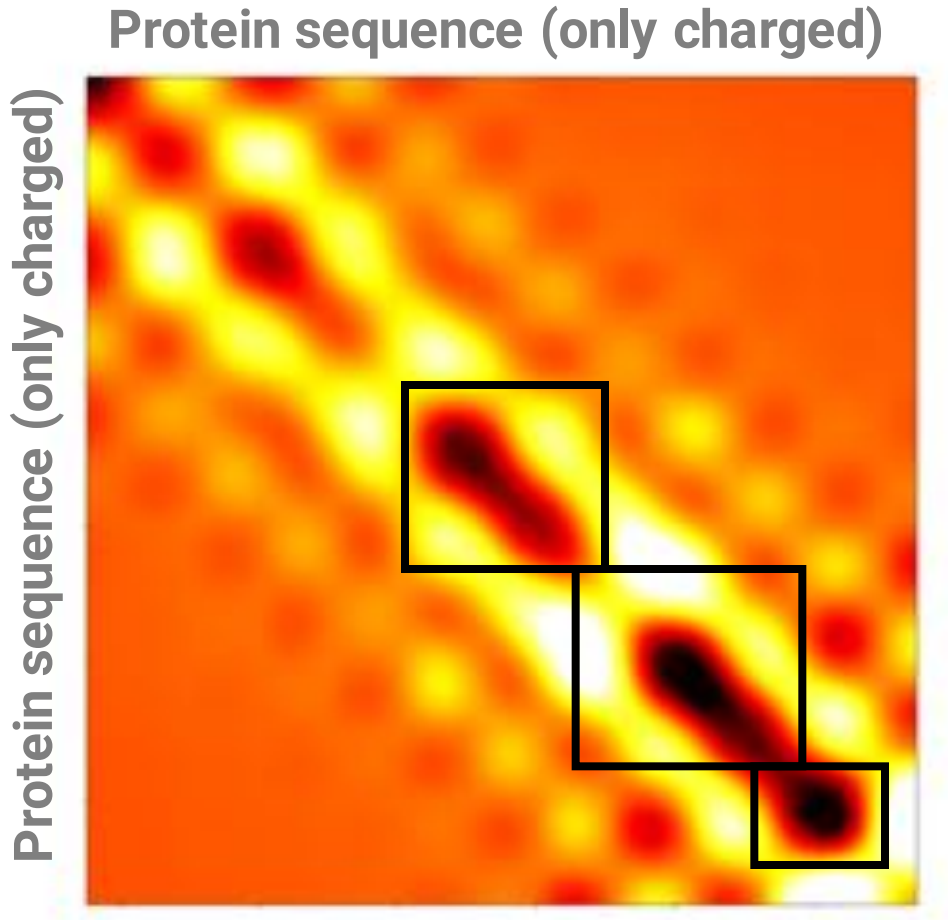
_peptone

Select residues

FVNQHLCGSHLVEALYLVCG

ERGFFYTPKTA

GIVEQCCTSICSLYQLENYCN



_peptone

20^{17} Only charged allowed → 5^{5}

Total number of mutants to screen

_peptone

Charge affects stability

Protein stability engineering via introduction of residual charges

_peptone

Identify pathological mutants



Journal List › J Clin Invest › v.118(6); 2008 Jun 2 › PMC2350430

JCI The Journal of Clinical Investigation

Published by The American Society for Clinical Investigation | Founded 1908

J Clin Invest. 2008 Jun 2; 118(6): 2148–2156. PMCID: PMC2350430

Published online 2008 May 1. doi: 10.1172/JCI33777

Seven mutations in the human insulin gene linked to permanent neonatal/infancy-onset diabetes mellitus

FVNQHLCGSHLVEALYLVCG

ERGFFYTPKTA



Journal List › J Diabetes Investig › v.2(2); 2011 Apr 7 › PMC4015537

JDI Journal of Diabetes Investigation

Official Journal of the Asian Association for the Study of Diabetes

J Diabetes Investig. 2011 Apr 7; 2(2): 92–100. PMCID: PMC4015537

Published online 2011 Feb 17. doi: 10.1111/j.2040-1124.2011.00100.x

Insulin gene mutations and diabetes

Masahiro Nishi[1] and Kishio Nanjo[*2]

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GIVEQCCTSICSLYQLENYCN

_peptone

5^5 Only charged allowed 5^3

Total number of mutants to screen

_peptone

FVNQHLCGSHLVEALYLVCG**E**RGFFYT**K**TA
VEQCCTSICSLYQL**E**NYCN

Humalog®

FVNQHLCGSHLVEALYLVCGERGFFYT**KP**TA
VEQCCTSICSLYQL**E**NYCN

_peptone

790.000x

30 years in 20 min.

_peptone

AI can relate vaguely connected physical concepts

AI and protein biophysics can be made complementary

AI can vastly accelerate combinatorial protein design

AI will significantly reduce the costs of protein therapeutics

AI will be the driving force behind synthetic biology

_peptone



Emanuele Paci
Chief Science Officer



Matt Heberling
Chief Operations Officer



Jan Domanski
Chief Technical Officer

Aarhus Uni.



Frans Mulder
Aarhus University

Google



François Chollet
Google AI

NVIDIA



Carlo Ruiz



Alison Lowndens



Mark Berger



Serge Lemonde

Oxbridge

UNIVERSITY OF OXFORD

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Thank you.

https://peptone.io